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Milestone 2: ML Solution DataViz	
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Program and Specialization	*BSIT Data Analytics*

I. Project Overview

FinMark Corporation has three datasets containing information about customer demographics, customer transactions, and social media interactions. Analyzing these datasets can help the company better understand its customers, transaction behavior, and online engagement. However, reviewing tables of data alone can make it difficult to recognize patterns, compare categories, and identify unusual values. Data visualization helps address this problem by presenting information through charts that are easier to understand and interpret.

The purpose of this milestone is to visualize and analyze the finalized datasets using Python. The team created histograms, count plots, a box plot, and a yearly transaction chart to examine numerical distributions, category frequencies, and time-based patterns. These visualizations support the exploratory data analysis (EDA) findings by presenting the data in a visual format.

The objectives of this milestone are to identify important patterns in the three datasets, assess the distribution and quality of the available data, compare visualization results with the findings from Milestone 1, and identify considerations for future machine learning development. The results will help the team understand the strengths and limitations of the datasets before selecting relevant features and developing a suitable machine learning solution for FinMark Corporation.

II. Visualization Overview

The team created visualizations for the three finalized datasets: Customer Demographics, Customer Transactions, and Social Media Interactions. The visualizations focus on customer characteristics, transaction patterns, and social media engagement. Different chart types were selected based on the type of information being analyzed. Histograms were used to examine numerical distributions, count plots were used to compare categorical frequencies, a box plot was used to inspect the spread of transaction amounts and potential outliers, and a yearly transaction chart was used to compare recorded transactions across years.

These visualizations were developed using Python libraries, particularly Pandas, NumPy, Matplotlib, and Seaborn. The results provide a visual summary of the data and help the team identify patterns, possible data-quality concerns, and areas that may require further investigation before machine learning model development.

A. Visualization Objectives

The visualizations were created to examine the distribution of customer ages and transaction amounts, compare the frequencies of categorical variables, and identify patterns in transaction records over time. The Customer Demographics dataset was used to examine age, gender, and income level. The Customer Transactions dataset was used to analyze transaction amounts, payment methods, product categories, and yearly transaction counts. The Social Media Interactions dataset was used to compare platforms and sentiment categories.

These variables were prioritized because they represent different aspects of customer characteristics and behavior. Histograms were used to examine numerical distributions, count plots were used to compare categorical frequencies, and a box plot was used to review the spread of transaction amounts and check for potential outliers. The yearly transaction chart was used to compare the number of recorded transactions across years.

The visualizations are important because they help the team identify patterns that may be relevant to future machine learning tasks. They also provide a visual way to assess data

quality, understand category representation, and recognize limitations that could affect later analysis.

B. Visualization Tools and Workflow

The activity used Python and the following libraries:

- **Pandas** was used to load, organize, and inspect the finalized datasets.
- **NumPy** was used to calculate numerical summaries such as the mean, median, standard deviation, variance, minimum, and maximum.
- **Matplotlib** was used to create and format the figures, including chart sizes, titles, axis labels, and display settings.
- **Seaborn** was used to create statistical visualizations, including histograms and count plots.

The finalized datasets were checked for missing values and unusual values. The Customer Demographics dataset contains 3,023 records, with no missing values and ages ranging from 18 to 70 years. The Customer Transactions dataset contains 3,015 records, with no missing amounts or negative amounts; however, 60 transactions have an amount of zero. The Social Media Interactions dataset contains 3,020 records, with no missing values, although some records are classified as “Unknown.” The available finalized datasets show the resulting values but do not independently confirm the exact cleaning and imputation procedures used in Milestone 1. These limitations should be considered when interpreting the visualizations.

III. Visualization Review and Analysis

A. Distribution and Frequency Analysis

The customer age histogram shows that ages range from 18 to 70 years, with a mean of approximately 44.69 and a median of 45. Age 45 appears 427 times, noticeably more often than any other individual age. This concentration is an observation in the finalized dataset; however, the available files do not establish its cause. The histogram shows that the dataset includes customers across a broad adult age range.

The gender count plot shows relatively similar numbers of female and male customers, although female customers are slightly more numerous. This suggests that neither category overwhelmingly dominates the gender distribution. The income-level chart also shows relatively similar frequencies among the High, Low, and Medium categories. The Unknown category has substantially fewer records than these named categories.

The transaction amount histogram shows that amounts range from 0.00 to 999.86, with a mean of approximately 493.62 and a median of 498.015. The finalized dataset contains 323 transaction amounts equal to 498.015. This concentration should be considered when interpreting the distribution, although the available finalized file alone cannot establish whether these repeated values resulted from median imputation or another cause.

The payment method count plot shows relatively similar numbers of transactions across the categories. Debit Card has the highest recorded frequency, while Bank Transfer has the lowest. The product category chart also shows relatively similar frequencies among the named categories, with the Unknown category having considerably fewer records.

For social media interactions, Instagram, Twitter, and Facebook have relatively similar recorded frequencies. The sentiment count plot shows that Positive, Neutral, and Negative are the most frequent sentiment categories, while Very Negative and Very Positive occur much less frequently. The Unknown category is also present and should be considered during future data preparation.

Overall, the frequency and distribution charts show that many categories have relatively similar representation, while some categories, especially Unknown and the two extreme sentiment categories, occur less frequently. These differences should be considered when preparing the datasets for machine learning.

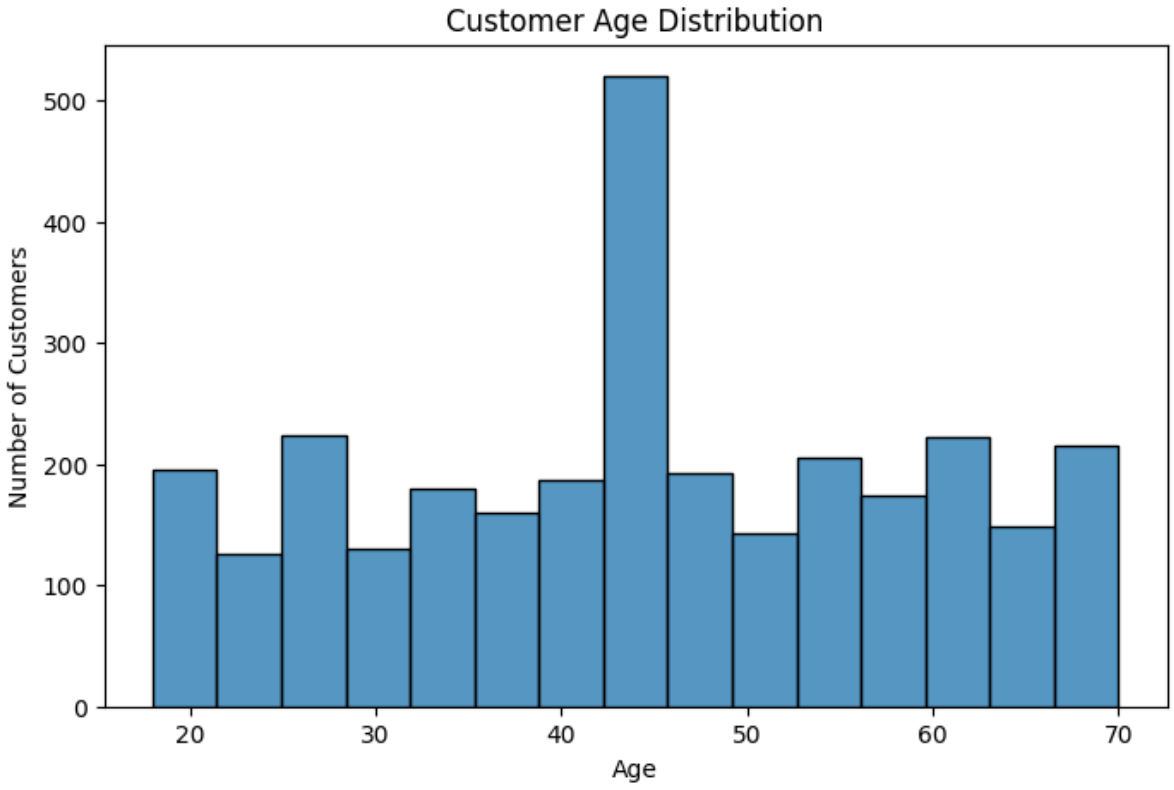


Figure 1. Customer Age Distribution

The customer age histogram shows that ages range from 18 to 70 years. The mean age is approximately 44.69, and the median is 45. The age of 45 appears 427 times, creating a noticeable concentration at this value. Although this pattern may warrant further investigation, the available finalized dataset does not establish why age 45 occurs so frequently. The distribution should therefore be interpreted alongside the limitations of the available preprocessing documentation.

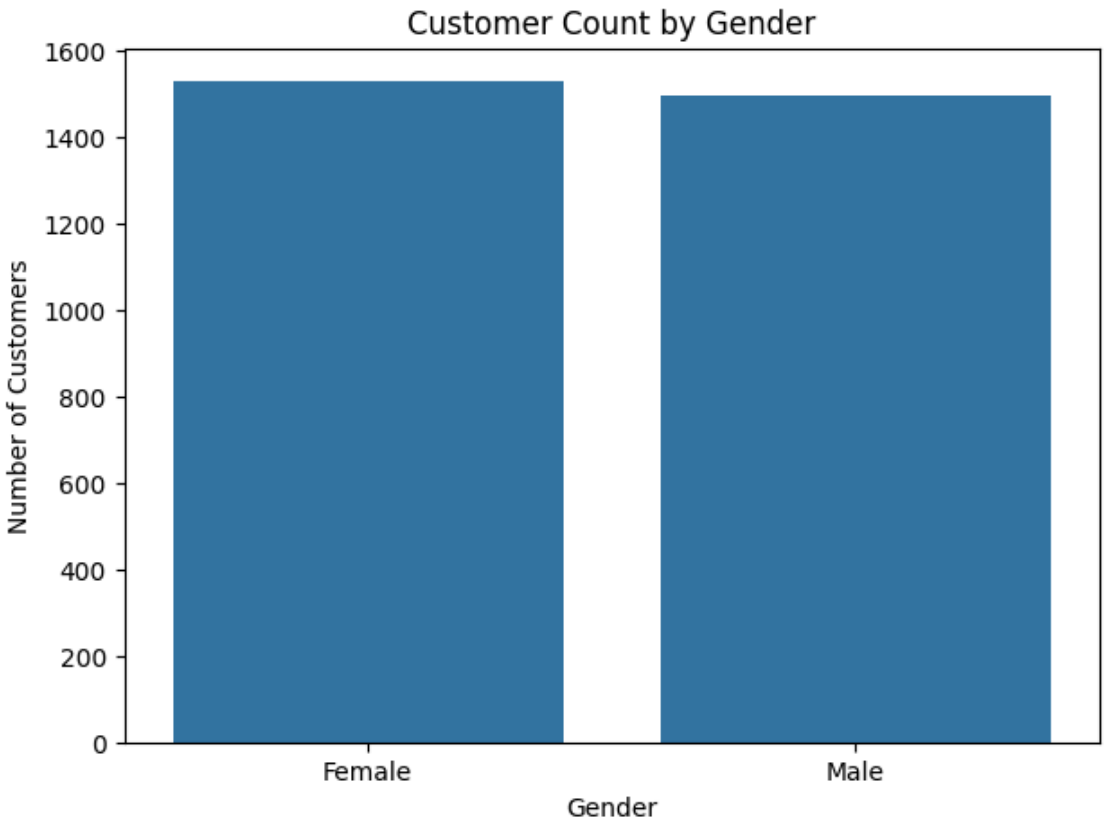


Figure 2. Customer Count by Gender

The gender count plot shows relatively similar numbers of female and male customers, with female customers slightly more numerous. This suggests that both categories are represented in the dataset without one category overwhelmingly dominating the other.

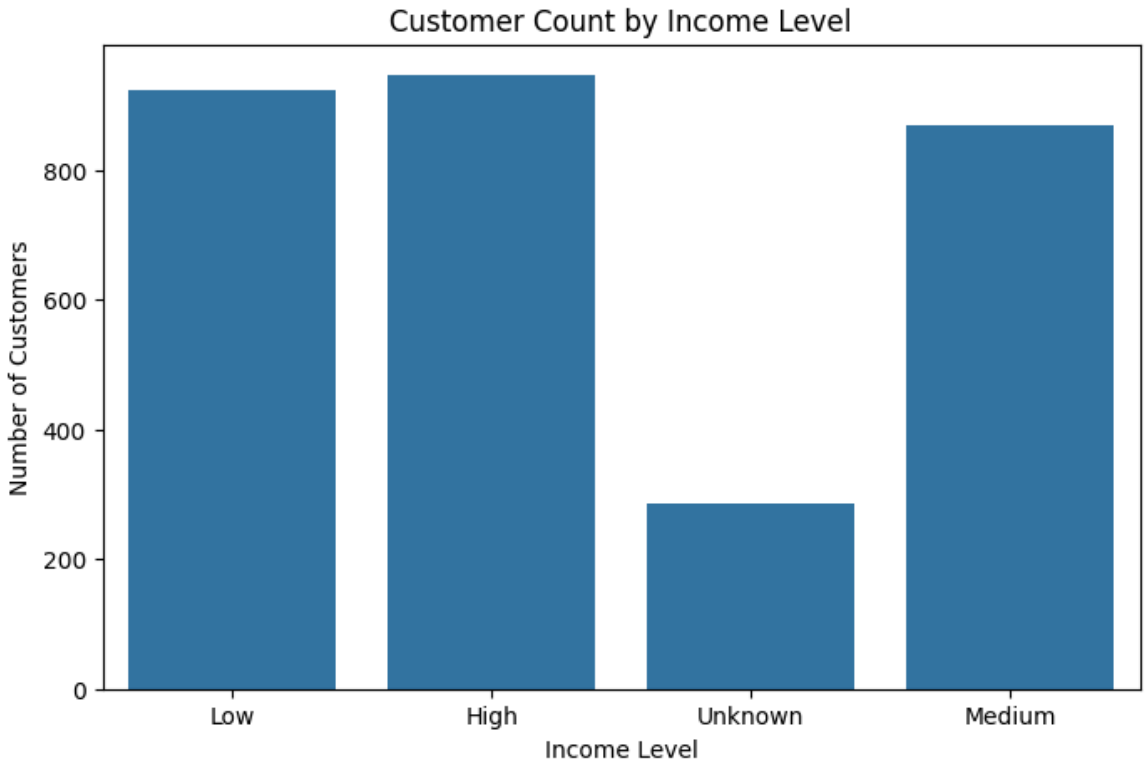


Figure 3. Customer Count by Income Level

The income-level count plot shows that the High, Low, and Medium categories have relatively similar frequencies, while the Unknown category has substantially fewer records. The team should consider the meaning of Unknown when preparing the data for future analysis.

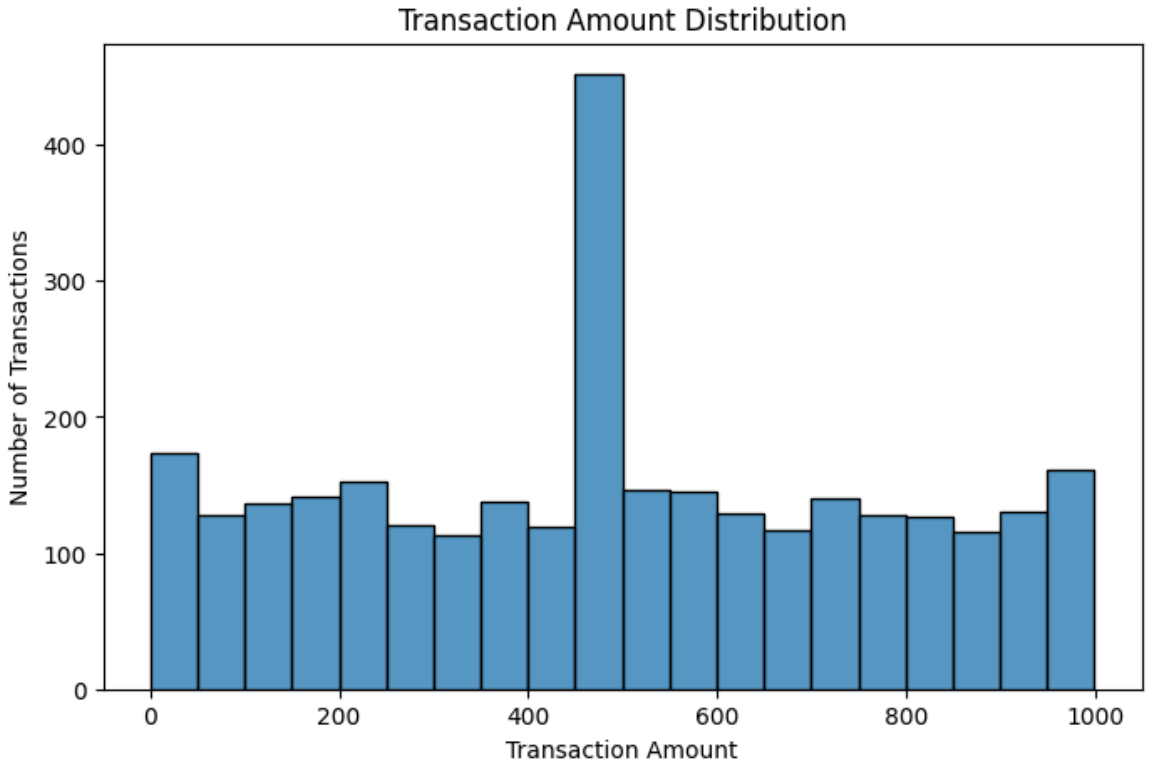


Figure 4. Transaction Amount Distribution

The transaction amount histogram shows values ranging from 0.00 to 999.86. The mean transaction amount is approximately 493.62, while the median is 498.015. The finalized dataset contains 323 records with an amount equal to the median value. This concentration is relevant when interpreting the distribution, but the available finalized dataset does not establish whether median imputation caused it. The 60 zero-amount transactions should also be reviewed against FinMark's business rules to determine whether they represent valid transactions.

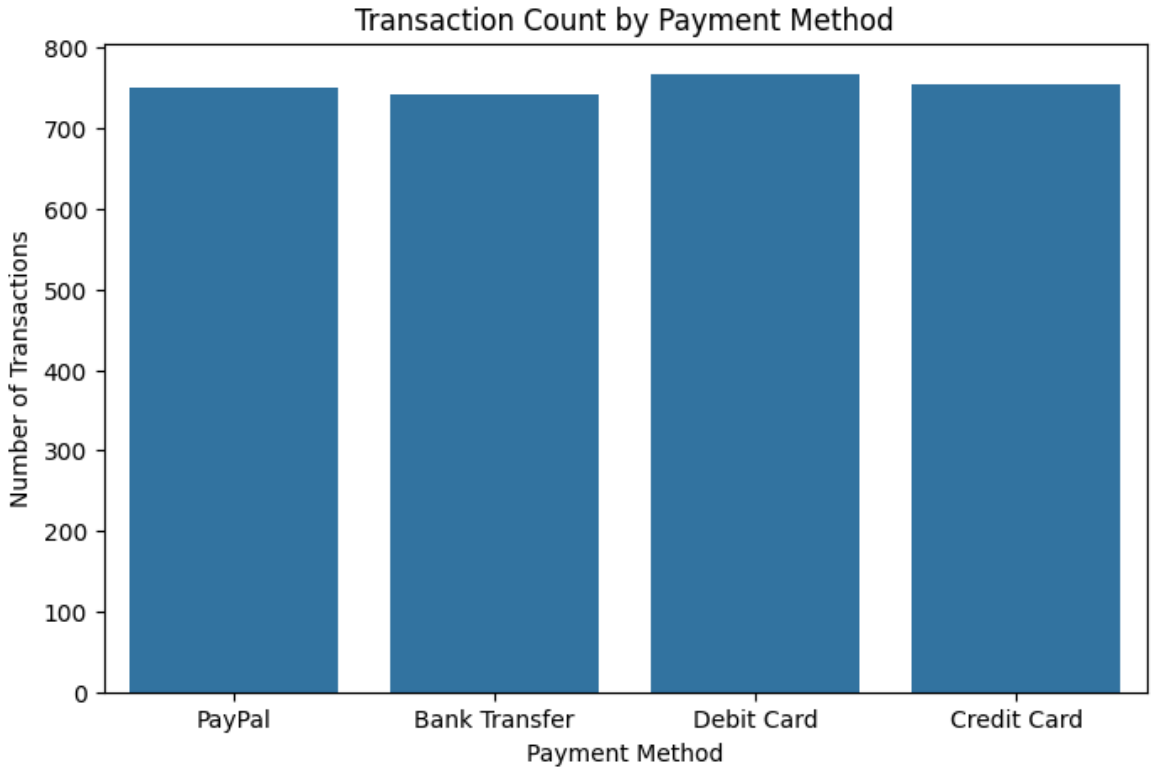


Figure 5. Transaction Count by Payment Method

The payment method count plot shows relatively similar frequencies across the recorded payment methods. Debit Card has the highest recorded frequency, while Bank Transfer has the lowest. These differences describe the distribution of payment methods in the available data but do not establish why customers use one method more often than another.

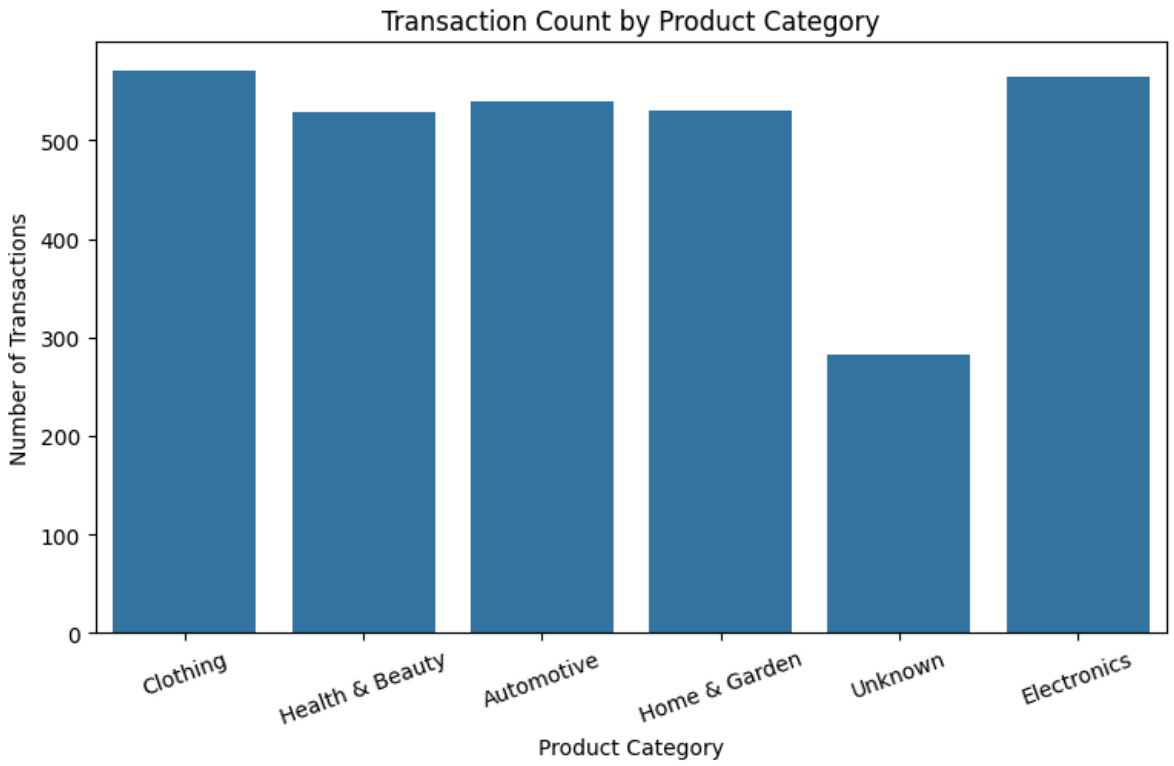


Figure 6. Transaction Count by Product Category

The product category count plot shows relatively similar frequencies among the named categories, while Unknown has substantially fewer records. This pattern suggests that the main product categories are represented in the dataset, but the Unknown category may require additional review.

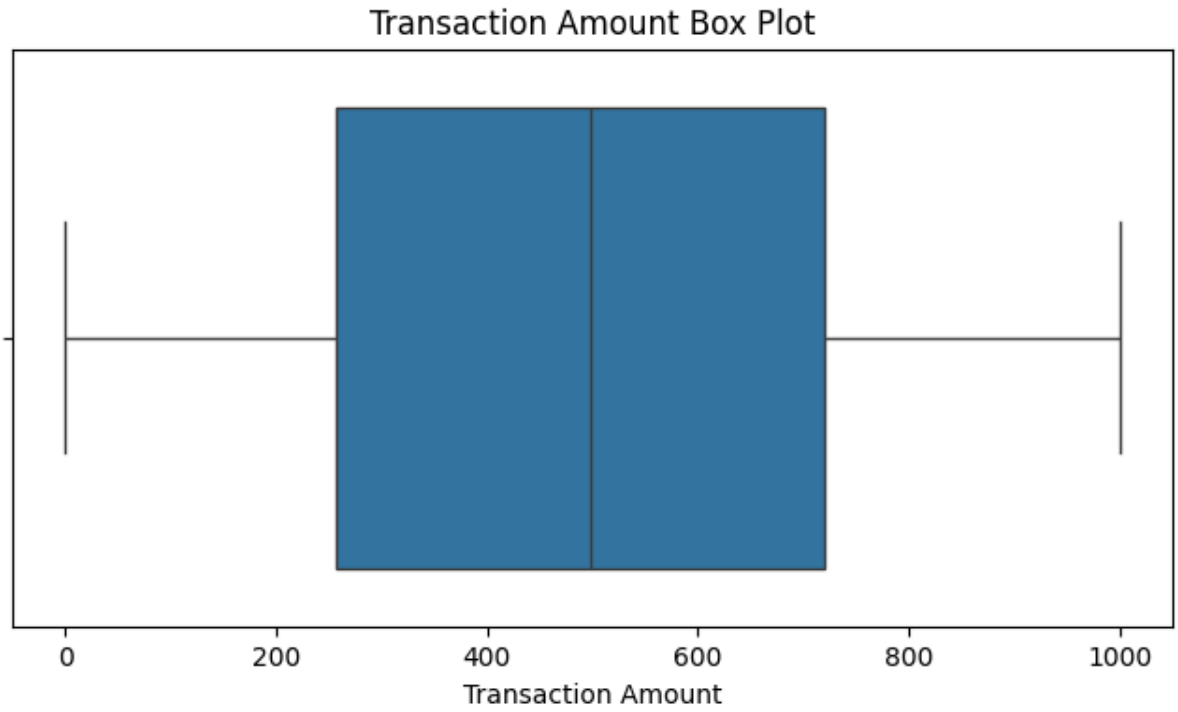


Figure 7. Transaction Amount Box Plot

The box plot summarizes the spread of transaction amounts and does not show obvious outliers beyond its whiskers. However, this does not guarantee that all values are valid. For example, a transaction amount of zero should be checked against FinMark's business rules to determine whether it represents a legitimate transaction.

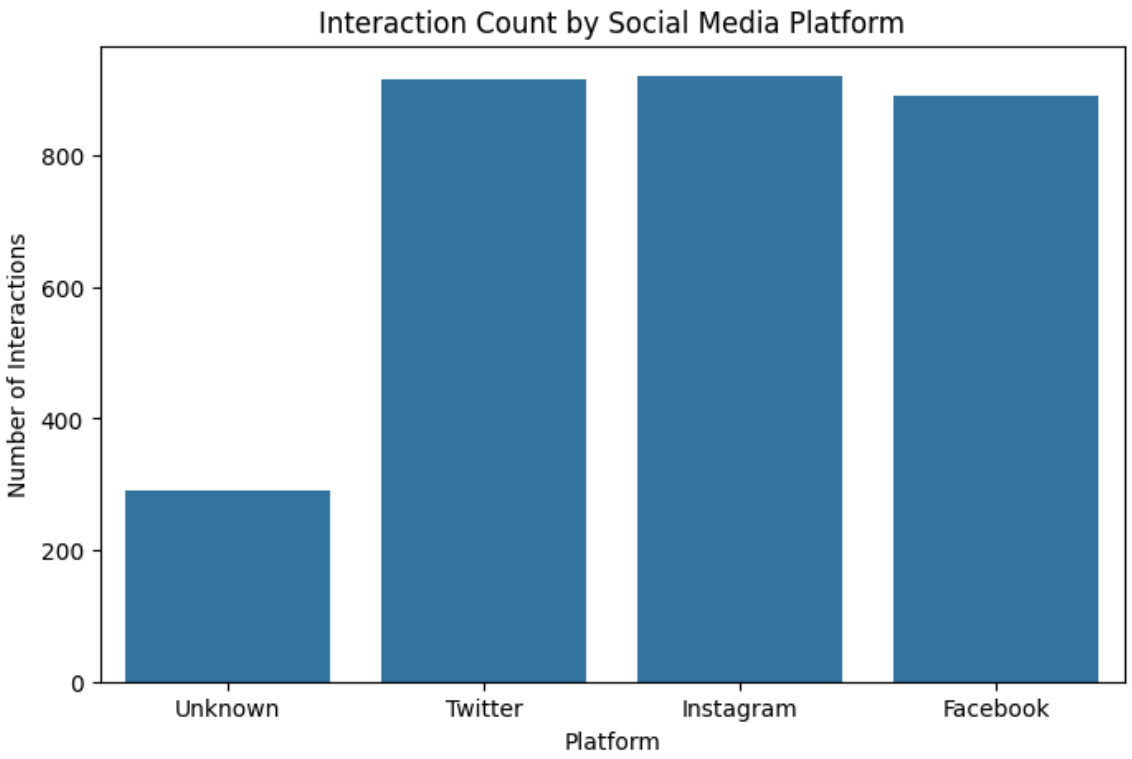


Figure 8. Interaction Count by Social Media Platform

The social media platform count plot shows relatively similar recorded interaction frequencies for Instagram, Twitter, and Facebook. This provides an overview of how the recorded interactions are distributed across the three platforms.

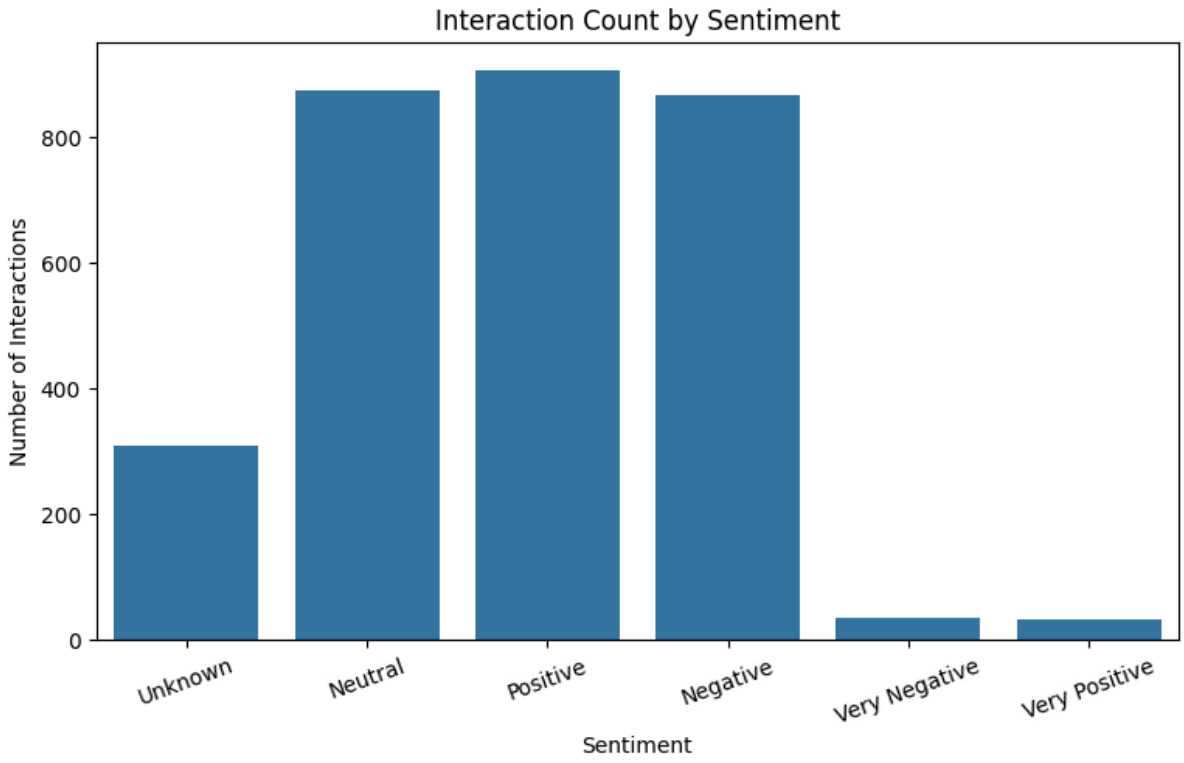


Figure 9. Interaction Count by Sentiment

The sentiment count plot shows that Positive, Neutral, and Negative interactions are more frequent than Very Negative and Very Positive interactions. The Unknown category is also present in the dataset, indicating that some sentiment records do not have a specified classification. The less frequent sentiment categories should be considered during future data preparation because a machine learning model may have fewer examples from which to learn patterns for these groups. The team should also review how Unknown sentiment values will be handled to ensure that preprocessing decisions are appropriate for the selected machine learning task.

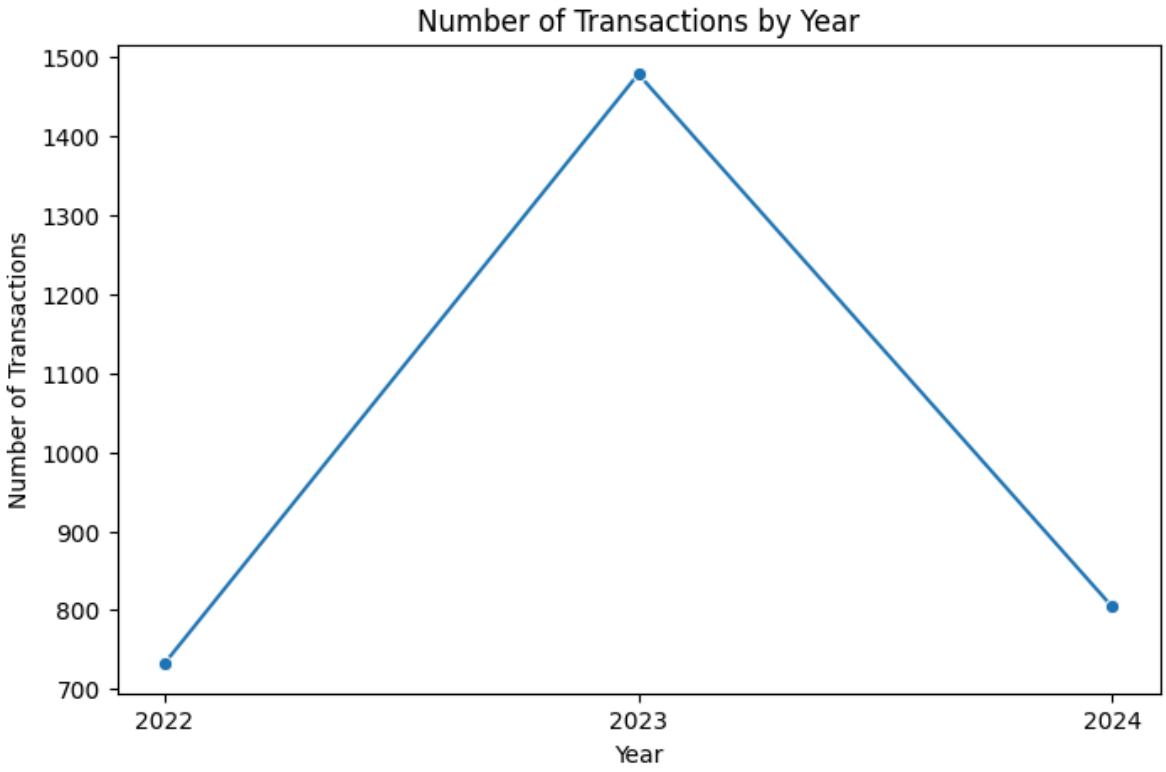


Figure 10. Number of Transactions by Year

The yearly transaction chart shows that 2023 has the highest number of recorded transactions, while 2022 and 2024 have fewer. However, the dataset has uneven monthly coverage across these years. Therefore, the differences in yearly totals should not automatically be interpreted as an increase or decrease in actual business activity.

B. Relationship and Trend Analysis

The current visualizations primarily examine individual-variable distributions and category frequencies rather than direct relationships between pairs of variables. They provide useful descriptive information, but additional visualizations or statistical methods would be needed to

investigate relationships between customer demographics and transaction behavior or between social media platforms and sentiment.

The age histogram, gender count plot, and income-level chart describe different aspects of the customer dataset. However, these visualizations do not establish whether age, gender, and income level are related to one another.

The yearly transaction chart shows that 2023 has the highest recorded transaction count. However, the dataset has uneven monthly coverage across the years, so differences in annual totals may reflect differences in the available observation periods rather than actual changes in business activity.

The social media charts show the distribution of recorded interactions across platforms and sentiment categories. The lower frequencies of Very Negative and Very Positive sentiments may be relevant to future data preparation, but the current charts do not establish an association between platform and sentiment.

For future analysis, the team could create grouped bar charts or other appropriate visualizations to examine relationships between transaction categories, customer characteristics, and social media sentiment. Suitable statistical methods could then be used to assess whether observed patterns represent meaningful relationships.

C. Outliers and Anomalies

The transaction amount box plot does not show obvious outliers beyond its whiskers. Nevertheless, the team should not assume that all transaction values are valid based on this visualization alone. Values such as zero may require further investigation to determine whether they represent legitimate transactions or possible data issues.

The transaction amount histogram shows a concentration of values around 500, including 323 records with an amount equal to 498.015. The finalized dataset alone does not establish the reason for this concentration or whether it resulted from median imputation. This pattern should be investigated further because repeated values can affect the interpretation of the distribution and may influence subsequent machine learning analysis.

The lower frequencies of the Unknown income and product categories, as well as the Very Negative and Very Positive sentiment categories, are additional observations to consider. These categories are not necessarily errors, but their lower representation may affect how well a model learns patterns associated with them.

Finally, the yearly transaction chart must be interpreted in light of the uneven monthly coverage across years. Before drawing conclusions about transaction trends, the team should check the available time periods and determine whether comparisons are fair. These observations highlight the importance of validating unusual values, understanding preprocessing decisions, and reviewing category representation before model development.

IV. Relationship Between Visualizations and Milestone 1 Findings

The visualizations support and clarify several findings from Milestone 1. The customer age histogram agrees with the descriptive statistics, which show a mean age of approximately 44.69 and a median of 45. The gender and income-level count plots also support the earlier frequency analysis, showing relatively similar counts across the main categories and fewer records in the Unknown income category.

The transaction amount histogram provides a visual representation of the numerical statistics, including a mean of approximately 493.62 and a median of 498.015. The box plot complements these results by showing the spread of transaction amounts and revealing no obvious outliers beyond its whiskers. However, the finalized dataset contains 60 zero-amount transactions and 323 records equal to 498.015. These observations should be considered

when interpreting the distribution, and the reason for the repeated median value cannot be confirmed from the finalized file alone.

The payment method, product category, social media platform, and sentiment charts also support the earlier findings about category frequencies. They show relatively similar frequencies among several common categories, while some categories, including Unknown and the two extreme sentiment categories, have fewer records.

The yearly transaction chart adds a visual comparison of transaction counts across years. It shows that 2023 has the highest recorded count, but uneven monthly coverage means the chart cannot conclusively establish a change in actual business activity.

Overall, the visualizations provide additional support for the descriptive findings from Milestone 1 and make the results easier to interpret. They also identify limitations that should be considered in future analysis, particularly the concentration of repeated values in the transaction dataset, the representation of less frequent categories, and differences in time coverage. The exact effects of the original cleaning and imputation procedures cannot be established from the finalized datasets alone.

V. Visualization Insights and Interpretation

The visualizations provide a clearer understanding of FinMark Corporation's customer demographics, transaction behavior, and social media interactions. Customer ages range from 18 to 70, with the mean and median both around 45 years old. The gender and income-level charts show relatively similar frequencies across the main categories, helping the team understand the composition of the customer dataset.

Transaction amounts range from 0.00 to 999.86, with a mean of approximately 493.62 and a median of 498.015. The finalized dataset contains 60 zero-amount transactions and 323 records equal to 498.015. Although the box plot shows no obvious outliers beyond its whiskers, this does not guarantee that every transaction value is valid. The zero amounts should be checked against FinMark's business rules, and the concentration around 498.015 should be investigated without assuming that it was caused by imputation.

The payment method chart shows that Debit Card has the highest recorded frequency and Bank Transfer has the lowest. The yearly chart shows the highest recorded transaction count in 2023, but uneven monthly coverage limits comparisons between years. In the social media dataset, Instagram, Twitter, and Facebook have relatively similar interaction frequencies, while Positive, Neutral, and Negative sentiments occur more frequently than Very Negative and Very Positive sentiments.

Age and Transaction Amount are numerical variables that may be considered for future machine learning tasks. Gender, Location, Income Level, Product Category, Payment Method, Platform, InteractionType, and Sentiment are categorical variables that may be relevant depending on the selected business problem. SignupDate, TransactionDate, and InteractionDate may provide date-related features if transformed appropriately. CustomerID, TransactionID, and InteractionID should primarily be treated as identifiers for linking or tracking records rather than automatically used as predictive features. The final selection of input features and any target variable should be guided by the prediction objective and evaluated through appropriate analysis and model testing.

For the next stage, the team should examine relationships between variables, review the meaning of zero transaction amounts, investigate the concentration of transaction amounts equal to 498.015, and determine how to handle low-frequency categories. The team should also account for uneven time coverage if it conducts further trend analysis. Reviewing the original cleaning records, if they become available, would help clarify how missing or invalid values were handled.

In conclusion, the visualizations help turn the EDA results into understandable patterns and provide a foundation for additional data analysis. By considering both the findings and the limitations, the team can make more informed decisions about feature selection, preprocessing, and the development and evaluation of a suitable machine learning solution for FinMark Corporation.

VI. Project Resources and Supporting Files

The following table lists the supporting datasets, code, and visualization outputs used in this milestone. These resources provide access to the materials supporting the team's data analysis and visualization results.

Reference No.	Reference	Link
1	Final Cleaned Dataset	Link
2	Python Code for Exploratory Data Analysis	Link
3	Python Code for Data Visualizations	Link
4	Generated Visualization Outputs	Link